

# Homework 1

Haoyang Sun

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## Abstract

This is my abstract!

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## 1. QUESTION 1

### 1.a. Proof

Start by taking logs on both sides of the return expression:

$$\log R_{e,t+1} = \log(1 + \exp(\log D_{t+1} - \log P_{t+1})) + \log P_{t+1} - \log P_t \quad (1)$$

For simplicity, I denote  $r_{e,t+1} = \log R_{e,t+1}$ ,  $p_{t+1} = \log P_{t+1}$ ,  $d_{t+1} = \log D_{t+1}$ , and  $dp_{t+1} = \log D_{t+1} - \log P_{t+1}$ , then, by a Taylor expansion in  $dp_{t+1}$  (which is stationary with mean  $\bar{dp}$ ), we get

$$r_{e,t+1} = \log(1 + \exp(\bar{dp})) + \frac{\exp(\bar{dp})}{1 + \exp(\bar{dp})} (dp_{t+1} - \bar{dp}) + p_{t+1} - p_t. \quad (2)$$

Denote  $\kappa = 1/(1 + \exp(\bar{dp}))$ , and  $\kappa_0 = \log(1 + \exp(\bar{dp})) - \bar{dp} \exp(\bar{dp})/(1 + \exp(\bar{dp})) = -\log \kappa - (1 - \kappa) \log(1/\kappa - 1)$ . The log-linearization becomes

$$r_{e,t+1} = \kappa_0 + (1 - \kappa) dp_{t+1} + p_{t+1} - p_t. \quad (3)$$

Moving  $p_t$  to the left-hand side and the other terms to the right-hand side, we get

$$\begin{aligned} p_t &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa p_{t+1} \\ &= \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa [\kappa_0 + (1 - \kappa) d_{t+2} - r_{e,t+2} + \kappa p_{t+2}] \\ &= \kappa_0 (1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + (1 - \kappa) (d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\ &\quad - [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1} r_{e,t+H}] \\ &\quad + \kappa^H p_{t+H} \end{aligned} \quad (4)$$

The log dividend-price ratio  $dp_t$  can then be written as

$$\begin{aligned}
dp_t &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1} d_{t+H}) \\
&\quad - \kappa^H p_{t+H} \\
&= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\
&\quad + [r_{e,t+1} + \kappa r_{e,t+1} + \dots + \kappa^{H-1} r_{e,t+H}] \\
&\quad d_t - d_{t+1} + \kappa(d_{t+1} - d_{t+2}) + \dots + \kappa^{H-1}(d_{t+H-1} - d_{t+H}) \\
&\quad + \kappa^H d_{t+H} - \kappa^H p_{t+H} \\
&= \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}
\end{aligned} \tag{5}$$

where  $\Delta d_{t+1} := d_{t+1} - d_t$ . Notice that the identity above holds ex post (state by state), so it also holds in conditional expectation at time  $t$ , giving

$$dp_t = \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} + \kappa^H \mathbb{E}_t dp_{t+H} \tag{6}$$

Letting  $H \rightarrow \infty$  and imposing the no-bubble (transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t dp_{t+H} = 0 \tag{7}$$

we obtain equation (1.3):

$$dp_t = \kappa_0 \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \tag{8}$$

1.b.

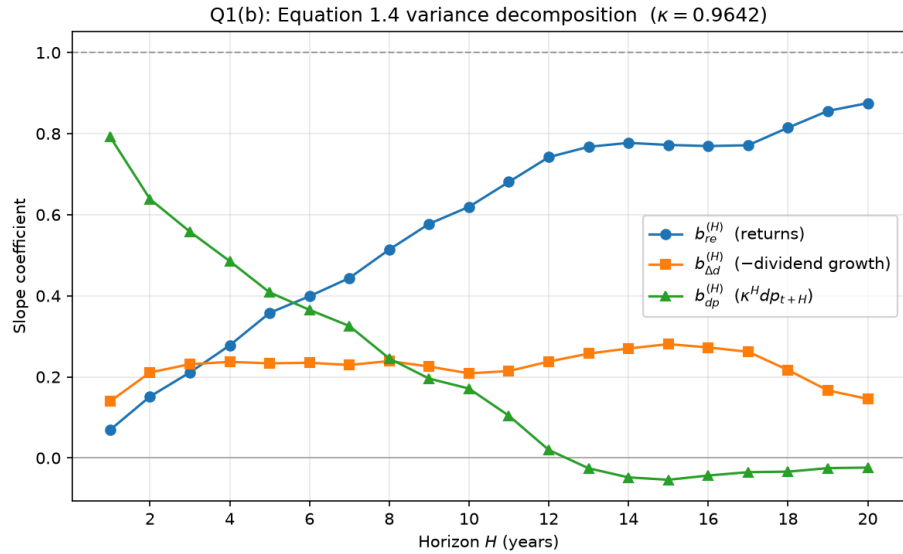


Figure 1: Equation 1.4 variance-decomposition slopes vs. horizon  $H$  (EQ Dataset, 1928–2021; overlapping monthly starts;  $\kappa = 1/(1 + e^{\bar{dp}}) = 0.9642$ ).

We can infer from the figure that, valuation ratio  $dp_t$  can predict log run return, and the volatility of price mainly comes from the volatility of discount rate.

1.c. VAR

Using equation

$$b_{re}^{(H)} = 1_{re}(\Gamma + \kappa^H \Gamma^{H+1})(I - \kappa\Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (9)$$

$$b_{\Delta d}^{(H)} = -1_{\Delta d}(\Gamma + \kappa^H \Gamma^{H+1})(I - \kappa\Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (10)$$

and  $b_{dp}^{(H)} = 1 - b_{re}^{(H)} - b_{\Delta d}^{(H)}$ .

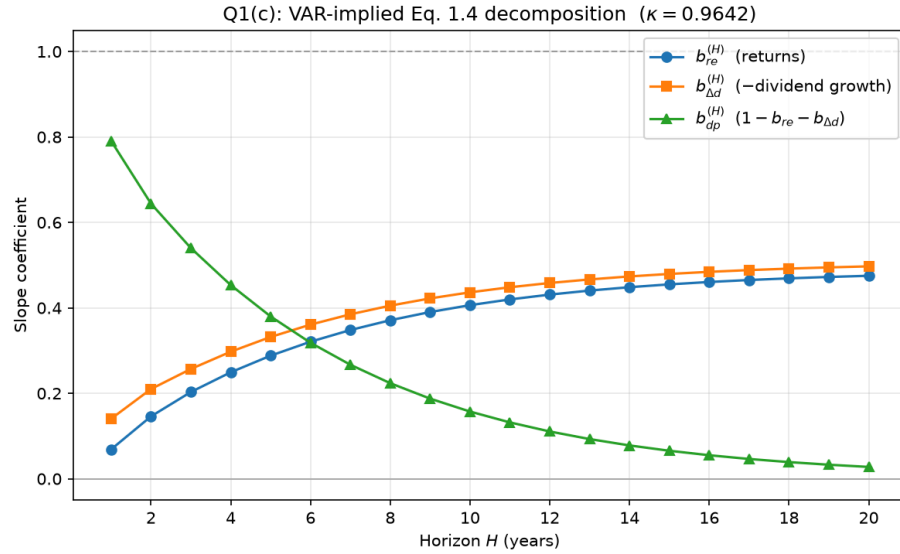


Figure 2: VAR-implied Equation 1.4 variance-decomposition terms vs. horizon  $H$ . Overlapping-annual VAR(1) on  $Z_t = [\Delta d_t, r_{e,t}, dp_t]'$  estimated by OLS (EQ Dataset, 1928–2021;  $\kappa = 0.9642$ , as in 1(b)).  $b_{dp}^{(H)}$  is imposed as  $1 - b_{re}^{(H)} - b_{\Delta d}^{(H)}$ .

1.d.

As  $H \rightarrow \infty$ , the

$$b_{re}^{(\infty)} = 1_{re}\Gamma(I - \kappa\Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (11)$$

$$b_{\Delta d}^{(\infty)} = -1_{\Delta d}\Gamma(I - \kappa\Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (12)$$

The calculated numbers are  $b_{re}^{(\infty)} = 0.4892$ ,  $b_{\Delta d}^{(\infty)} = 0.5096$ , and  $b_{dp}^{(\infty)} = 0.0012$ . These numbers show that valuation ratio  $dp$  can both predict dividend growth and returns, and half of price volatility is attribute to dividend growth variance, half is attribute to discount rate variance. This result is not aligned with Cochrane(2011).

1.e. Alternative log-linear present-value identity

Using the same trick,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \frac{D_{t+1}}{D_t} \frac{D_t}{P_t + D_t} \frac{P_t + D_t}{P_t} \quad (13)$$

, the log one-period return can be written as

$$r_{e,t+1} = -\log(1 - e^{-dy_{t+1}}) + \Delta d_{t+1} + \log(1 - e^{-dy_t}) + dy_t \quad (14)$$

where  $dy_t := \log\left(1 + \frac{D_t}{P_t}\right)$  and  $\log(1 - e^{-dy_t}) = \log \frac{D_t}{P_t + D_t}$ .

Applying a Taylor expansion of  $\log(1 - e^{-dy_t})$  in  $dy_t$  around its mean  $\bar{dy}$ , we have

$$\log(1 - e^{-dy_t}) = \log(1 - e^{-\bar{dy}}) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} (dy_t - \bar{dy}) + O\left((dy_t - \bar{dy})^2\right) \quad (15)$$

Letting  $\kappa = e^{-\bar{dy}}$  to simplify the expression,

$$\log(1 - e^{-dy_t}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + O\left((dy_t - \bar{dy})^2\right) \quad (16)$$

Then the return is log-linearized as

$$r_{e,t+1} = -\frac{\kappa}{1 - \kappa} (dy_{t+1} - \bar{dy}) + \Delta d_{t+1} + \frac{\kappa}{1 - \kappa} (dy_t - \bar{dy}) + dy_t \quad (17)$$

and

$$\begin{aligned} \frac{dy_t}{1 - \kappa} &= r_{e,t+1} - \Delta d_{t+1} + \kappa \frac{dy_{t+1}}{1 - \kappa} \\ &= r_{e,t+1} - \Delta d_{t+1} \\ &\quad + \kappa \left[ r_{e,t+2} - \Delta d_{t+2} + \kappa \frac{dy_{t+2}}{1 - \kappa} \right] \\ &= r_{e,t+1} - \Delta d_{t+1} + \kappa [r_{e,t+2} - \Delta d_{t+2}] \\ &\quad + \dots + \kappa^{H-1} [r_{e,t+H} - \Delta d_{t+H}] + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \\ &= \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \end{aligned} \quad (18)$$

Thus, the log-linearization identity is

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H} \quad (19)$$

Taking the conditional expectation at time  $t$ , the identity also holds

$$dy_t = (1 - \kappa) \left( \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \right) + \kappa^H \mathbb{E}_t dy_{t+H} \quad (20)$$

Imposing the no-bubble assumption, the last term vanishes as  $h \rightarrow \infty$ ,

$$\lim_{H \rightarrow \infty} \kappa^H E_t \log\left(\frac{P_t + D_t}{P_t}\right) \rightarrow 0 \quad (21)$$

## 2. QUESTION 2

2.a.

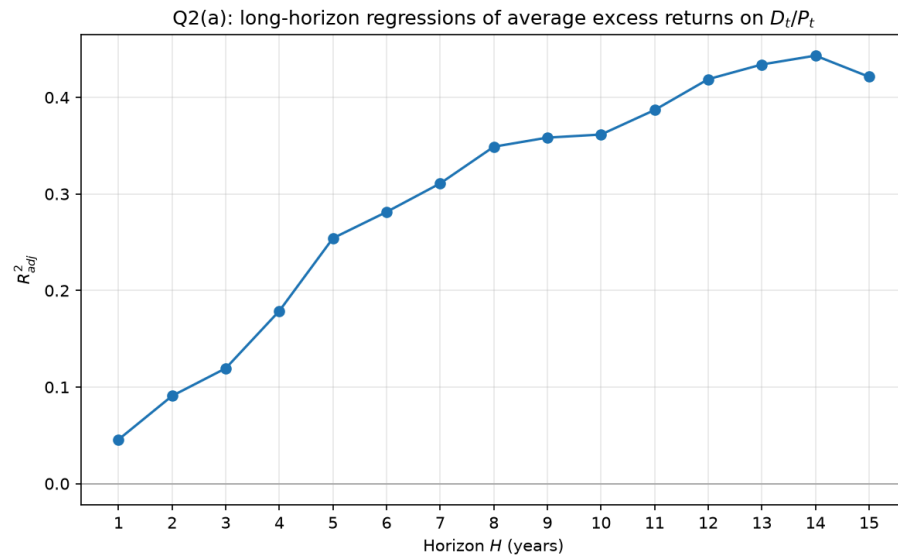


Figure 3: Adjusted  $R^2$  of the Equation 2.1 regressions of the average future excess equity return  $\frac{1}{H} \sum_{h=1}^H x R_{e,t+h}$  on  $D_t/P_t$ , plotted against the horizon  $H = 1, \dots, 15$  years (EQ Dataset, 1927:12–2021:12; overlapping monthly start dates, with year  $h$  taken 12h months ahead, so horizon  $H$  uses  $N - 12H$  observations).

Valuation ratio  $\frac{D}{P}$  did better job in predicting long run return.

2.b.